$\boxed{01/21/2026}$

... assumptions

Thm [Newton's Method convergence]: let $x \in [a,b]$ be a zero of $f \in C^2[a,b]$ ... $x_n \to x^*$ at order 2.

pf: By Taylor Expansion

$$f(x) = f(x_k) + f'(x_k)(x - x_k) + f''(\xi_k)(x - x_k)^2$$

$$0 = f(x^*) = f(x_k) + f'(x_k)(x^* - x_k) + f''(\xi_k)(x^* - x_k)^2$$

$$\Rightarrow \quad f'(x - x_k)(x_k - x^*) - f(x_k) = \frac{1}{2} f''(\xi_k)(x^* - x_k)^2$$

@ order 2 means

$$\lim_{n\to\infty} \frac{e_{k+1}}{e_k^2} \quad \lim_{n\to\infty} \frac{|x_{k+1} - x^*|}{|x_k - x^*|^2} = C < \infty.$$

$$x_{k+1} - x^* = \left(x_k - \frac{f(x_k)}{f'(x_k)}\right) - x^* \qquad \text{by N.M.}$$

$$= \underbrace{(x_k - x^*)}_{e_k} - \frac{f(x_k)}{f'(x_k)}$$

$$= \frac{f'(x_k)(x_k - x^*) - f(x_k)}{f'(x_k)} = \frac{f''(\xi_k)(x^* - x_k)^2}{f'(x_k)}$$

$$\Rightarrow e_{k+1} = \frac{f''(\xi_k)(x^* - x_k)^2}{2f'(x_k)} \Rightarrow \frac{e_{k+1}}{e_k^2} = \frac{\|f''(\xi_k)\|}{2\|f'(x_k)\|}$$

$$\lim_{k\to\infty} \frac{e_{k+1}}{e_k^2} = \lim_{k\to\infty} \frac{\| f''(\xi_k)\|}{2\|f'(x_k)\|} < \infty$$

$\therefore$ By N.M we get conv. by order 2.

<u>Main limitation</u> $f(x)$ can be unknown

<u>Sol</u>: Instead of $f'(x_n)$ find some g_n s.t. $g_n \approx f'(x_n)$
g_n easier to compute

$$\boxed{\begin{array}{l} \text{Quasi-Newton Method} \\ x_n = x_{n-1} - \dfrac{f(x_{n-1})}{g_{n-1}} \qquad n \geq 1 \end{array}}$$

$$\boxed{\begin{array}{l} \text{Secant Method} \\ x_{n+1} = x_n - \dfrac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n), \qquad n \geq 1 \end{array}}$$

Draw Secant

Newton's Method for systems of Nonlinear Eq.

$$\vec{x}_k = \vec{x}_{k-1} - \vec{F}'(x_{k-1})^{-1}\vec{F}(x_{k-1}), \quad \text{for} \quad k \geq 1$$

$$\vec{F}(x) = \left(\frac{\partial f}{\partial x_j}\right)_{ij}$$

$$f(x,y) = 2x + xy - 1 = 0 \qquad \vec{x}_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$
$$g(x,y) = 2y - xy + 1 = 0$$

$$\Rightarrow \begin{pmatrix} f_1(\vec{x}) \\ \vdots \\ f_n(\vec{x}) \end{pmatrix} = \begin{pmatrix} f_1(\vec{x}_k) \\ \vdots \end{pmatrix} + \underbrace{\begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f}{\partial x_k} \\ \vdots & & \\ \frac{\partial f_n}{\partial x_1} & & \end{pmatrix}\Bigg|_{\vec{x}_k}}_{\vec{F}'(\vec{x}_k)} (\vec{x} - \vec{x}_k) + \underbrace{h.o.t.}_{\text{higher order terms}}$$

N.M.

$$\Rightarrow \boxed{\vec{F}(\vec{x}) = \vec{F}(\vec{x}_k) + \vec{F}'(\vec{x}_k)(\vec{x} - \vec{x}_k)} + h.o.t.$$

<u>Ex</u> $f(x,y) = 2x + xy - 1 = 0$ $\qquad 2\times 2$

$g(x,y) = 2y - xy + 1 = 0$

2×1 ??

$$\begin{pmatrix} 2x + xy - 1 \\ 2y - xy + 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \qquad \vec{F}(\vec{x}) = \begin{pmatrix} 2+y & x \\ -y & 2-x \end{pmatrix}$$

$$\vec{x}_0 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\vec{x}_1 = \vec{x}_0 - \vec{F}'(\vec{x}_0)^{-1}\vec{F}(x_0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}^{-1} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix}$$

$$\vec{x}_2 = \begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix} - \vec{F}'\left(\begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix}\right)^{-1} \cdot \vec{F}\left(\begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix}\right) = \begin{pmatrix} 3/4 \\ -3/4 \end{pmatrix}$$

$$\vec{x}_2 = \begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix} - \begin{pmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{pmatrix}^{-1} \cdot \begin{pmatrix} -1/4 \\ 1/4 \end{pmatrix}$$

$$\vec{x}_3 = \begin{pmatrix} 0.875 \\ -0.875 \end{pmatrix}, \quad \vec{x}_k \to \vec{x}^* = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

<u>Thm</u> [Convergence of Newton's Method] Consider a nonlinear system $\vec{F}(\vec{x}) = \vec{0}$, $\vec{F} : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$. $\vec{F}$ satisfies the following 3 conditions:

- $\exists x^*$, s.t. $\vec{F}(\vec{x}^*) = \vec{0}$
- Jacobian $\vec{F}'(x^*)$ is non-singular
- Jacobian $\vec{F}' : \Omega \to \mathbb{R}^n$ is Lipschitz cont. w/ Lipschitz constant γ.
  i.e.

$$\|\vec{F}'(\vec{x}) - \vec{F}'(\vec{y})\| \leq \gamma \|\vec{x} - \vec{y}\|$$

$\exists\, k > 0$ and $\delta > 0$ s.t. $\forall \vec{x}_0 \in B_\delta(\vec{x}^*)$, the seq. $\{\vec{x}_k\}$ generated by Newton's Method

$$\vec{x}_k = \vec{x}_{k-1} - \vec{F}'(\vec{x}_{k-1})^{-1} \cdot \vec{F}(\vec{x}_{k-1}), \quad \text{for } k \geq 1$$

satisfies $\vec{x}_k \in B_\delta(\vec{x}^*)$, and

<u>proof</u> Consider the error, $\underbrace{\vec{x}_{k+1} - \vec{x}^*}_{e_{k+1}} = \underbrace{\vec{x}_k - \vec{x}^*}_{e_k} - \vec{F}'(\vec{x}_k)^{-1} \cdot \vec{F}(\vec{x}_k)$

$$\Rightarrow \vec{e}_{k+1} = \vec{F}'(\vec{x}_k)^{-1}\left[\vec{F}'(\vec{x}_k)(\vec{x}_k - \vec{x}^*) - \left[\vec{F}(\vec{x}_k) - \vec{F}(\vec{x}^*)\right]\right]$$

$$= \vec{F}'(\vec{x}_k)^{-1} \left[\int_0^1 \vec{F}(\vec{x}_k)(\vec{x}_k - \vec{x}^*)\,dt - \int_0^1 \vec{F}'(\vec{x}^* + t(\vec{x}_k - \vec{x}^*))(\vec{x}_k - \vec{x}^*)\,dt \right]$$

$$= \vec{F}'(\vec{x}_k)^{-1} \int_0^1 \left[\vec{F}'(\vec{x}_k) - \vec{F}'(x^* + t(\vec{x}_k - \vec{x}^*) \right] (\vec{x}_k - \vec{x}^*)\,dt$$

$$\Rightarrow \quad \|\vec{x}_{k+1} - \vec{x}^*\| \le \|\vec{F}'(\vec{x}_k)^{-1}\| \cdot \underbrace{\left\| \int_0^1 \left[F'(\vec{x}_k) - \vec{F}'(\vec{x}_k + t(\vec{x}_k - x^*) \right] (\vec{x}_k - \vec{x}^*)\,dt \right\|}_{I}$$

$$\le 2\|\vec{F}(\vec{x}^*)^{-1}\| \cdot I$$